



Honda ASIMO robot

that all this might never happen. The reason, in short, is that the technologies have not advanced sufficiently to make robotics mainstream.

There have been innumerable experimental successes to show that robots are indeed capable of replicating many aspects of human nature and replacing humans in many conventional roles. However, there is a lot more to be done before these technologies can be manifested as consumer products. If you have seen Rajnikanth's latest blockbuster *Enthiran*, *The Robot*, this point would need no further elaboration.

Fahad Azad, managing partner of Robosoft Systems, explains what could possibly be the reason for this: "Robots are still at a nascent stage and are learning to walk and move around unstructured environments. They are able to see, hear and touch but cannot make as much sense out of it like humans do. Robots outperform humans in controlled environment and automated manufacturing, though."

Inventions and experiments related to the sensory and perception skills of robots are especially crucial. Let us quickly review recent developments in this space.

## I, the robot, am no longer short-sighted!

The biggest advancements in robotics have been on the sensory front. Keeping the sixth sense aside, there has been major headway in trying to replicate all of the other senses.

"The robots making headlines today are mostly humanoid robots whose perception skills have vastly changed and improved from the initial robots made commercially-available. The most commonly known humanoid robot is Honda's ASIMO which is currently in its twelfth model, since the E0 in 1986. If we look back at the E0 and compare it with today's ASIMO, the changes are galaxies apart," says Praveen Pitchai, another of Dr Srinivasa's students.

"Sensory skills in robots have so vastly improved that sight and hearing are now a part and parcel of every humanoid robot. Robots like the Geminoid and the Actroid developed by Osaka University are changing the way we look at robot perception skills. Feeling and tasting are the two perceptions that are really difficult to simulate within a humanoid robot, but the Geminoid and the Actroid show

us that even feeling can be simulated. Artificial intelligence (AI) has played a major role in the development of these skills. Also, the hardware aspect of these robots has changed a lot. No longer are normal power sources being used; instead, alternative sources like compressed air and fuel cells are being used to power them. All in all, with the implementation of AI, alternative power, very large-scale integrated (VLSI) chips and advanced sensors, robotics is rapidly evolving to match human beings. The days of Isaac Asimov's 'I, Robot' are not too far away," he adds.

Indeed, the sensing powers of robots have increased manifold. At one time, most robots were 'bump-and-go.' That is, they would stop and turn only when they hit an obstacle. Instead, today's robots use a variety of cameras, infrared and ultrasound sensors, etc, along with computer vision and machine perception technologies that make sense of the gathered information. More advanced robots can even smell, taste and touch.

"In 2009, scientists in Illinois developed a small, inexpensive, lab-on-a-chip sensor that quickly and accurately identifies sweetness—one of the five primary tastes. It can identify with 100 per cent accuracy the full sweep of natural and artificial sweet substances, including 14 common sweeteners. In 2008, an electronic tongue capable of telling the differences between various wines was developed, thanks to half a dozen sensors that measure levels of acid, sugar and alcohol. Other such technologies have come up in recent years," explains Jayakrishnan.

The human capability of touch, or haptic feedback, has also been emulated in robots, enabling them to perform sensitive tasks like surgeries. One recent example is Sofie (Surgeon's Operating Force-feedback Interface Eindhoven)—the surgical robot.

The version of Sofie revealed in the last quarter of 2010 is equipped not just with a whole toolbox of surgical equipment and an Internet connection (that enables surgeons from anywhere in